

Jeremy Hsu

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EDUCATION

Pennsylvania State University

Bachelor of Science, Computer Science (College of Engineering)

University Park, PA

Aug. 2025 – May 2028

- **Relevant Coursework:** Applied Regression Analysis, Discrete Mathematics for Computer Science, Linear Algebra, Multivariable Calculus & Vector Analysis, Data Structures

EXPERIENCE

Strategy Developer (Evaluation)

Z22 Technologies AG

Feb. 2026 – Present

Zurich, Switzerland (remote)

- Selected by a Swiss quantitative asset manager to submit 3 algorithmic strategies to their evaluation track; contracted sub-advisory with profit share on deployment
- Identified and self-reported a lookahead bias in a candidate backtest before deployment, tracing it to an unshifted return window; adopted signal-delay decay testing as a validation check
- Deliver live allocation signals via REST API, implementing webhook submission directly in the NinjaTrader C# strategy to replace manual delivery for daily and intraday signals

Founder & Quantitative Researcher

Weaver Capital LP

Jan. 2026 – Present

West Chester, PA

- Built and deployed 2 uncorrelated algorithmic strategies in futures/equities, spanning intraday/swing/macro horizons
- Engineered execution infrastructure in Python and C# with automated strategy synchronization, enforced position and drawdown limits, atomic state persistence across restarts, and sub-second order handling
- Built portfolio optimization pipelines allocating capital dynamically across uncorrelated sleeves, sizing positions on realized/implied volatility and balancing risk at the portfolio level

Quantitative Research Lead

Penn State Quantitative Finance Club

Aug. 2025 – Present

University Park, PA

- Lead strategy research for the club's algorithmic paper portfolio, signal development, model validation, and execution
- Develop forecasting models for equity indices and commodities, focused on macroeconomic cycles and regime shifts

Undergraduate Researcher

Locomotion in Biology and Robotics (LiBR) Lab, Penn State

Aug. 2026 – Present

University Park, PA

- Built physics simulations of multi-body contact dynamics in MuJoCo, modeling an articulated three-section robot in confined geometry to evaluate gait strategies before hardware testing
- Modeling scaling laws for computational versus mechanical cost of transport under Dr. Baxi Chong

PROJECTS

Rime — deployed strategy, Weaver Capital LP | *Python, pandas, NumPy, Alpaca API, Streamlit*

2025 – Present

- Built a systematic daily allocator rotating leveraged Nasdaq ETFs, a volatility hedge, and cash via a shallow regime tree, a cross-asset macro overlay, and a volatility circuit-breaker with hysteresis
- Unified research and production behind a single decision function with persisted state, so live orders cannot diverge from the backtest; data-quality checks and a rebalance deadband to reduce turnover
- Validated via walk-forward folds, pre/post-2020 splits, slippage/commission stress, and block bootstrap; 47 pytest tests over the signal tree and risk overlays
- Simulated 2012–2025 at ~44% average gross exposure: Sharpe 1.86 vs. 0.92 for SPY, max drawdown –17% vs. –34%; Sharpe 1.58 excluding 2020. Live on Alpaca since Apr. 2026

Fifth Sanctuary — deployed strategy, Weaver Capital LP | *Python, C#/NinjaScript, MQL5*

2025 – Present

- Built an intraday gold volatility-expansion system with independent macro-scaled long and short risk sleeves
- Filtered the short sleeve out of the macro regimes where research showed it was structurally unfavorable; shipped a lookahead-safe Python macro script (FRED/Yahoo, T+1 shifted) feeding daily sleeve weights to the live C# strategy
- Simulated on spot XAU hourly bars, 2004–2025: PF 1.48, Sharpe 1.38, ~18% time in market. Live since Jan. 2026

TECHNICAL SKILLS

Languages: Python, C++, C#, Java, Pine Script, SQL

ML / Statistics: scikit-learn, Statsmodels, walk-forward validation, Monte Carlo, block bootstrap

Data Analysis: pandas, NumPy, SciPy, TA-Lib, Matplotlib, Plotly, Parquet

Platforms & Tools: Alpaca API, NinjaTrader, QuantConnect, Backtrader, Streamlit, pytest, Git, Linux